22102124 MIS

USN | | | | | | | | | | [22TCSBS122]

FOURTH SEMESTER B.Tech EXAMINATION, FEB - 2024

ENVIRONMENT AND ECOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	What is an ecosystem?	L-1	CO1	**02**
	b.	Define biodiversity hotspot.	L-2	CO2	**02**
	c.	Write any two differences between renewable and non-renewable energy sources.	L-1	CO3	**02**
	d.	Describe noise pollution.	L-2	CO4	**02**
	e.	What is a catalytic converter?	L-1	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Write in detail about grass land ecosystem.	L-1	CO1	**05**
		OR			
	b.	Discuss the thermodynamic laws involved in the energy flow in ecosystem with suitable examples.	L-2	CO1	**05**
3.	**a.**	Summarize the salient feature of hotspots of biodiversity.	L-3	CO2	**05**
		OR			
	b.	List out the bio-geographical zones of India.	L-2	CO2	**05**
4.	**a.**	Narrate any five uses of natural resources.	L-2	CO3	**05**
		OR			
	b.	Describe the key functions of forest resources.	L-2	CO3	**05**
5.	**a.**	Define acid rain and write the chemical reactions involved.	L-2	CO4	**05**
		OR			
	b.	Explain the photochemical smog in detail.	L-2	CO4	**05**
6.	**a.**	Illustrate the concept of ozone layer depletion.	L-3	CO5	**05**
		OR			
	b.	Explain the ways of achieving the sustainable development.	L-2	CO5	**05**
7.	**a.**	Discuss the biodiversity at regional or local level.	L-2	CO2	**05**
		OR			
	b.	Write a short note on lithosphere.	L-3	CO1	**05**

		PART-C	**06 X 10=60**		
8.	**a.**	Elaborate the structure and functions of the aquatic ecosystem.	L-3	CO1	**10**
		OR			
	b.	Describe atmosphere and hydrosphere.	L-2	CO1	**10**
9.	**a.**	Narrate the strategies, advantages and disadvantages involved in ex-situ conservation methods.	L-2	CO2	**10**
		OR			
	b.	Discuss in-situ approach for the biodiversity conservation in detail.	L-2	CO2	**10**
10.	**a.**	i. What are the uses of forest natural resources? ii. How forest natural resources are over exploited?	L-1	CO3	**10**
		OR			
	b.	Define water conservation. What are the three important measures for water conservation?	L-2	CO3	**10**
11.	**a.**	Summarize the radioactive pollution.	L-3	CO4	**10**
		OR			
	b.	Write in detail about waste-water treatment with a neat diagram.	L-1	CO4	**10**
12.	**a.**	Elaborate the preparedness and response methods involved in disaster management.	L-3	CO5	**10**
		OR			
	b.	Summarize the positive and negative impacts of environment on human health.	L-3	CO5	**10**
13.	**a.**	Interpret the various threats to biodiversity.	L-3	CO2	**10**
		OR			
	b.	Explain the five important natural resources and their uses.	L-3	CO3	**10**